



Inspiring Excellence

BRAC UNIVERSITY

CSE360: Computer Interfacing

Lab Project Report

Title: *Pet Health monitoring and automated feeding system*

by

Group_No 4
Section 8

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Abstract

This project presents an IoT-based Pet Health Monitoring and Automated Feeding System designed to assist busy pet owners in maintaining their pets' well-being. The system includes an Arduino Uno microcontroller to enable automated food dispensation, environmental monitoring and tracking pet activity. The arduino is programmed to operate servo motors for the scheduled feeding and the ambient temperature and humidity are monitored by the DHT11 sensor, which, if the conditions become unsafe, will activate a DC fan, too. In addition, infrared (IR) sensors are used to detect the presence of the pet, and alert the pet via a Bluetooth module and a buzzer if the pet does not feed within a 24-hour period. These findings provide a reliable, cost-effective and energy efficient solution that decreases food waste and alleviates owner stress, thereby supporting sustainable and accessible pet care.

Keywords

microcontroller, automation, IoT, pet-health, embedded systems

1. Introduction

With busy work schedules, it can be difficult for many pet owners to keep an eye on their pet's diet or environmental conditions. Extended leave may result in poorly scheduled feeding and failure to note illnesses, as well as dangerous environmental conditions. This project aims to build a smart and automated system for monitoring and taking care of the general health of pets like birds, cats and dogs at home using a remote system.

This project will cover elements such as autonomous feeding, real-time environmental monitoring (temperature/humidity), activity monitoring around the feeding station and sending real-time updates to the owner's smartphone through Bluetooth. This project is very relevant to Computer Interfacing because it involves connecting the microcontroller (Arduino Uno) with a number of analog and digital sensors, actuators, and communication protocols to form a responsive embedded system.

2. Related Work/ Inspiration

The conventional automatic pet feeders are mainly mechanical timers and do not have environmental monitoring or health tracking capabilities. Smart feeders are available commercially and have Wi-Fi cameras and app connection, but are usually costly and cannot control the immediate surroundings of the pet.

This project was inspired by the need for a complete and low cost solution. It sets itself apart with its environmental control (the fan automatically starts when the temperature becomes too hot) and safety-check algorithms (24 hours no food intake alert system with IR sensors and 24 hours no water intake alert system with Water-level sensor). This all-encompassing solution goes beyond mere feeding to help maintain the pet's comfort and notify the owner about possible health issues (such as sudden loss of appetite). Along with helping with the hygiene, by ensuring overflow occurs through stopping the dispenser when the food bowl is full.

3. Technical Approach

System Architecture:

The Arduino Uno serves as the central processing unit.

- **Inputs:**DHT11 (environment), IR sensors (food intake and level tracking) and Water level sensor(water tracking)
- **Outputs:** Servo motor (dispenser lid), DC Fan (cooling), LCD (local display), Buzzer (local alarm), and Bluetooth Module (remote alerts).

Components Used:

- **Arduino Uno:** The core microcontroller processing sensor data and controlling actuators.
- **Sensors:** IR Sensors (detects pet presence), DHT11 (measures temperature/humidity), Water Level Sensor.
- **Actuators & Outputs:** Servo Motors (rotates 90° to open/close food containers), DC Fan (cools the environment), LCD Monitor (displays system status locally), Buzzer (audible alerts).
- **Communication:** Bluetooth Module (transmits data to the owner's smartphone).

Cost Breakdown:

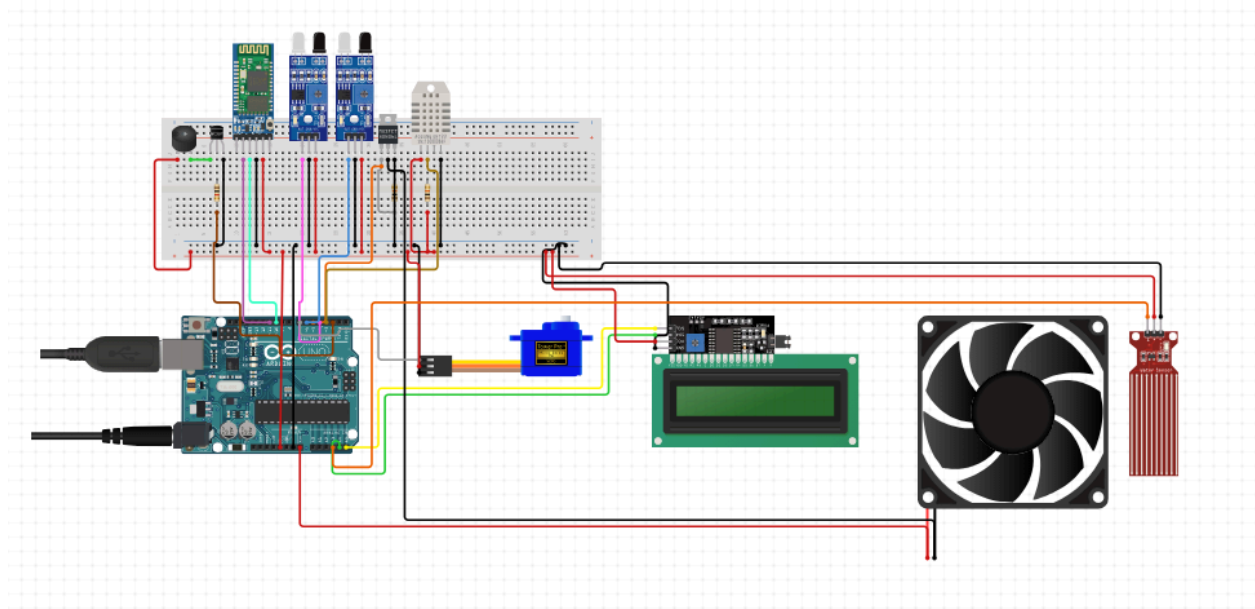
Component	Unit Cost (BDT)	Quantity	Total (BDT)
Arduino Uno	690	1	690
IR sensor	45	2	90
Servo motor	143	2	286
DC fan	110	1	110
LCD Monitor	195	1	195

Bluetooth module	349	1	349
Temp & Humidity (DHT11)	120	1	120
Buzzer	15	1	15
Breadboard	90	1	90
5V Power Adapter	160	1	160
Water level sensor	49	1	49
Total Approximate Cost			2,154

Communication Protocols:

1. **UART/Serial Communication:** This is used by the Bluetooth module to send the alerts to the smartphone app, and during debugging by the Serial Monitor.
2. **PWM (Pulse Width Modulation):** Used to precisely set the servo motor's angular position to rotate 90° to feed the food (90° angle) and to control the speed of the DC fan.
3. **I2C (Inter-Integrated Circuit):** This is used for the LCD monitor, thus reducing the number of wires used to only SDA and SCL lines.

Circuit/Schematic:[circuit/schematic](#)



Implementation Steps

1. **Time-Based Feeding:** A timer is running on the Arduino constantly. It sends a PWM signal to the servo motor at specific intervals, opening the dispenser for a specific amount of time only, to avoid overfeeding.
2. **Environment Regulation:** The DHT11 polls ambient conditions. If thresholds are exceeded, the DC fan is activated (For now, we used the humidity threshold which is easier to control and better for demonstration where exceeding 75% humidity turns the fan on) the bluetooth and LCD will show the humidity and temperature live. Besides, the Fan status is also displayed (Serving as a warning that the threshold is crossed).
3. **Inactivity Alert:** A timer is constantly running in the background. If the IR sensor detects the pet, the timer resets. If it reaches 24 hours without an IR trigger, the buzzer sounds and a Bluetooth alert is pushed to the owner.

Challenges:

The biggest hurdle was the power consumption: The power was reduced when the servo, DC fan and the Bluetooth module were all turned on at once. This was solved by providing enough current for the power adapter, and by using decoupling capacitors on the breadboard. Another was finding a reliable supplier. Many of our components that we bought first were faulty (bluetooth module, DHT11, servo, etc). Finally, we had to purchase them once more. Causing a great deal of delay and problems for us.

4. Sustainability & Impact

Sustainability: It dispenses measured quantities of food, thus preventing food waste and food spoiling. High energy efficiency is achieved by the use of low power sensors and a microcontroller that only turns on the high power components when it is absolutely necessary, such as the DC fan. Additionally, the chassis can be constructed from re-used or recycled materials (Our own was constructed from the cardboard box that our components had come in).

Impact: This system will greatly ease the mind of working pet owners, decreasing their anxiety about the health of their pet. It can detect behavioral changes at an early stage (e.g., 24-hour anorexia), which can help avoid severe health crises, resulting in improved animal welfare.

Future Work: Future versions could involve Wi-Fi connectivity for worldwide surveillance (IoT over MQTT), camera modules for visual verification, and solar panels to run the system off-grid.

Limitations: If the owner is away from home, they will not get immediate push notifications as it is based on Bluetooth and it has a limited range (usually 10 meters). Further, a loss of power will cause the system to be inoperable without the addition of a battery back-up module.

5. Results & Discussion

The prototype was able to complete all the core goals. The servo motor was able to accurately turn 90° of rotation at the correct time and without jamming the food dispenser. The DHT11 sensor was able to successfully capture the changes in the environment; when a simulated humidity was introduced into the sensor, the DC fan immediately turned on and reduced the local humidity and temperature. In addition, 24-hour IR absence logic and 24-hour water level not decrease logic worked perfectly even in the case of a shortened time scale test (e.g., 1-minute interval for debugging), giving the buzzer and the smartphone's Bluetooth alert as well as the LCD correct warning.

6. Conclusion

The Pet Health Monitoring and Automated Feeding System is one such example to solve a real-world problem using interfacing with a microcontroller. The adoption of time-scheduled feeding, environmental control with DHT11 sensor, and IR based food volume monitoring ensured a successful implementation of the desired smart pet care system, achieving all the goals proposed by the project. Although this was technically successful, it has also been confirmed by the use of UART, PWM, and I2C protocols. Overall, the project highlights the feasibility of sustainable, automated solutions for pet care and their role in enhancing animal welfare and pet owners' satisfaction.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Md. Rafi Uddin Auntik	22201075	Hardware design, Circuit building, code,etc	● Contributed in the C code writing for various components ● Contributed in the circuit building ● Primary circuit design and component list making ● Report writing
Naved Abrar Nibir	22201025	Software Design,Component buying, code,etc	● Contributed in the C code writing for various components ● Contributed in the circuit building ● Bought the components for the project. ● Report writing
Md Saidul Islam Apu	21301668	Hardware Design,Component buying, code,etc	● Contributed in the C code writing for various components ● Contributed in the circuit building ● Report writing
Fariha Yasmin	22301331	Software Design,Hardware design,code,etc	● Contributed in the C code writing for various components ● Contributed in the circuit building ● Report writing

References

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